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waseemshahzad@Waseem-Sf x + v - □ ×
o-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-buil
tin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -f
no-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/printf.
o kernel/printf.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -
DSOL_SYSCALL -DLAB_SYSCALL -MD -mcmmodel=medany -ffreestanding -fno-common -nostdli
b -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memse
t -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fn
o-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-buil
tin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -f
no-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/uart.o
kernel/uart.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -
DSOL_SYSCALL -DLAB_SYSCALL -MD -mcmmodel=medany -ffreestanding -fno-common -nostdli
b -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memse
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tin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -f
no-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinloc
k.o kernel/spinlock.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel
kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/pr
oc.o kernel/switch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/syp
roc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kerne
l/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/vi
rtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/s
pinlock.o
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permiss
ions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /;
/^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/waseemshahzad/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (3.5s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.7s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.3s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.7s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.0s)
== Test attack ==
$ make qemu-gdb
attack: OK (1.3s)
== Test time ==
time: OK
Score: 45/45
waseemshahzad@Waseem-Shahzad:~/xv6-labs-2025$

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waseemshahzad@Waseem-Sh x waseemshahzad@Waseem-SI x + v
(gdb) num = p->trapframe->a7; ut outptimized out:133ll.c:133
119 [SYS_sbrk] sys_sbrk,
120 [SYS_pause] sys_pause,
121 [SYS_uptime] sys_uptime,
122 [SYS_open] sys_open,
123 [SYS_write] sys_write,
124 [SYS_mknod] sys_mknod,
125 [SYS_unlink] sys_unlink,
126 [SYS_link] sys_link,
127 [SYS_mkdir] sys_mkdir,
8H
;7LL am129 |;
92G^130m1m1m8G

3L;71;3L;7L132 | syscall
B+ 133 {
134 int num;
135 struct proc *p = myproc();
136
> 137 num = p->trapframe->a7;
138 if(num > 0 && num < NELEM(syscalls) && syscalls[num]) {
139 // Use num to lookup the system call function for num, call it,
140 // and store its return value in p->trapframe->a0
141 p->trapframe->a0 = syscalls[num]();
142 } else {
143 printf("%d %s: unknown sys call %d\n",
144 p->pid, p->name, num);
145 p->trapframe->a0 = -1;
146 }
147 }

2G||;1H;10;1;7L6LL
7 f0
pagetable = 0x87f52000, trapframe = 0x87f56000, context = {ra = 0x800012fc,
sp = 0x3fffffdc50, s0 = 0x3fffffdc80, s1 = 0x80007cb0, s2 = 0x80007880, s3 = 0x0,
s4 = 0x80012dc0, s5 = 0x3, s6 = 0x80018950, s7 = 0x0, s8 = 0x80018a78, s9 = 0x182,
s10 = 0xb0, s11 = 0x2}, ofile = {0x0 <repeats 16 times>}, cwd = 0x80015dc0, name = {
0x69, 0x6e, 0x69, 0x74, 0x0 <repeats 12 times>}}
(gdb) p /x *p
$3 = {lock = {locked = 0x0, name = 0x80007118, cpu = 0x0}, state = 0x4, chan = 0x0,
killed = 0x0, xstate = 0x0, pid = 0x1, parent = 0x0, kstack = 0x3fffffd000, sz = 0x5000,
pagetable = 0x87f52000, trapframe = 0x87f56000, context = {ra = 0x800012fc,
sp = 0x3fffffdc50, s0 = 0x3fffffdc80, s1 = 0x80007cb0, s2 = 0x80007880, s3 = 0x0,
s4 = 0x80012dc0, s5 = 0x3, s6 = 0x80018950, s7 = 0x0, s8 = 0x80018a78, s9 = 0x182,
s10 = 0xb0, s11 = 0x2}, ofile = {0x0 <repeats 16 times>}, cwd = 0x80015dc0, name = {
0x69, 0x6e, 0x69, 0x74, 0x0 <repeats 12 times>}}
(gdb) layout src
(gdb) backtrace
#0 syscall () at kernel/syscall.c:137
#1 0x0000000080001a9c in usertrap () at kernel/trap.c:68
#2 0x00000003ffffff09c in ?? ()
(gdb) |
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